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FedCLIP-Distill: Heterogeneous federated cross-modal knowledge distillation for multi-domain visual recognition

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ABSTRACT

Federated learning (FL) for multi-domain visual recognition confronts significant challenges due to heterogeneous data distributions and domain shifts, which severely impair the semantic generalization capability of existing methods. To address these challenges, we propose FedCLIP-Distill, a novel framework that employs dual-domain knowledge distillation (KD) and contrastive relational distillation (CRD) to leverage the powerful visual-language alignment of CLIP in heterogeneous FL environments. Our approach employs a centralized CLIP teacher model to distill robust visual-textual semantics into lightweight client-side student models, thereby enabling effective local domain adaptation. We provide a theoretical convergence analysis proving that our distillation mechanism effectively mitigates domain gaps and facilitates robust convergence under non-IID settings. Extensive experiments on Office-Caltech10 and DomainNet benchmarks show that FedCLIP-Distill outperforms other methods: it achieves an average cross-domain accuracy of 98.5% on Office-Caltech10 and 80.50% on DomainNet. In different heterogeneous situations (e.g., Dirichlet $\alpha = 0.5$, 9.52% higher than FedCLIP), demonstrating significant improvements in accuracy and generalization under heterogeneous scenarios. The source code is available at https://github.com/Yuankun-Xia/FedCLIP-Distill.

1. Introduction

With the rapid development of edge computing ecosystems, unstructured data generated by devices including Internet of Things (IoT) sensors, smartphones, and medical imaging equipment has entered a stage of explosive growth[1,2]. However, centralizing this data for model training raises significant privacy concerns and incurs substantial communication overheads [3]. The General Data Protection Regulation (GDPR) [4] for user-generated content strictly prohibits the transmission of sensitive visual data. For instance, sharing a patient's MRI image with a central server may violate patient privacy protection terms and regulatory compliance requirements. Moreover, the volume of edge vision data is huge, resulting in extremely high communication overhead. Transmitting such data to a central server will saturate network bandwidth and introduce unacceptable delays, rendering the deployment of real-time edge applications unfeasible [5,6].

FL has evolved into a transformative learning paradigm to address the aforementioned bottlenecks [7–9]. In the FL architecture, a central server coordinates multiple distributed clients to jointly train a shared global model: each client trains a local model using its private local data and only uploads model parameter updates to the server, which then iteratively optimizes the global model by aggregating these updates [10]. This architecture not only preserves data privacy by retaining sensitive visual data locally [11] but also significantly reduces communication costs. For example, uploading parameter updates of a lightweight visual model like MobileNetV2 [12] consumes merely megabytes of bandwidth, far less than the terabyte-scale bandwidth needed for raw data transmission. The utility of FL has been demonstrated across various computer vision tasks, such as image classification [13,14] and object detection [15,16].

Despite its promise, the practical real-world deployment of FL is hindered by fundamental challenges. A primary obstacle is data heterogeneity, characterized by client data that is non-IID [17–20]. This phenomenon stems from a combination of user behavior patterns, environmental conditions, and differences in edge device hardware. For instance, images captured by smartphones and those taken by laptops have completely different feature distributions. More importantly, the usage scenarios of the equipment have further exacerbated the skew of data distribution. For instance, images captured by low-resolution webcams typically exhibit significant noise and fixed viewing angles, whereas images taken by high-resolution cameras often feature varied lighting and complex backgrounds. Such discrepancies in data acquisition devices and styles result in severe feature distribution shifts, causing the data across clients to deviate significantly from the IID

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Fig. 1. Example of the domainnet dataset, which includes six domains.

assumption and thereby undermining the efficacy of global model aggregation. Compounding these difficulties is model heterogeneity, arising from disparate client computational resources and network capabilities. This diversity leads to variations in local model architectures and inconsistent update schedules [21], rendering the direct parameter aggregation employed by conventional algorithms (e.g., FedAvg [22]) ineffective or entirely infeasible.

In multi-domain visual recognition, these challenges are further amplified. Client data not only exhibits non-IID characteristics but also originates from distinct domains [23]. Taking the DomainNet dataset [24] as an example (as shown in Fig. 1), it includes 6 domains with distinctly different visual styles, covering 345 object categories. Images from each domain exhibit significant visual distribution differences due to variations in generation methods or collection scenarios. For instance, the same category "dog" appears completely different across domains [25]. Standard FL frameworks struggle to address this issue because they only focus on the parameter aggregation process and fail to effectively align cross-domain semantic representation, ultimately frustrating the collaborative learning goal of cross-domain generalization.

To address these interconnected challenges, we propose a novel FL framework, FedCLIP-Distill, that synergistically integrates a pre-trained Vision-Language Model (CLIP) [26] with KD [27]. Our approach is specifically designed to overcome the limitations of conventional FL frameworks in heterogeneous and multi-domain settings. By leveraging the powerful, generalizable representations of CLIP as a global teacher model, our method effectively mitigates the negative impacts of domain shifts. Simultaneously, this robust knowledge is transferred via KD to lightweight and potentially heterogeneous client models, thereby addressing both data and model heterogeneity. Our framework enables the training of a robust, generalizable model that achieves superior performance and adaptability across diverse clients and domains while preserving data privacy. The primary contributions of the paper are as follows:

- We design a federated framework where a CLIP model serves as a centralized teacher to encode visual-textual correspondences, while lightweight student models on each client adapt to local domain characteristics. This allows for cross-domain knowledge transfer via CLIP's shared semantic space, mitigating the impact of data heterogeneity.
- We introduce a domain-aware distillation loss to minimize inter-domain feature discrepancy using a domain classifier and a visual-language alignment regularization to enforce consistency between student-generated visual embeddings and CLIP's text embeddings. These components jointly enhance domain generalization and semantic consistency.
- Extensive experiments on Office-Caltech10 and DomainNet show that FedCLIP-Distill achieves higher cross-domain accuracy and stronger robustness against data heterogeneity compared to baseline methods, validating the effectiveness of our cross-modal FL approach.

The remainder of this paper is organized as follows. Section 2 presents related work. Sections 3 and 4 present the problem definition and system architecture. Section 5 provides a convergence analysis of our scheme and proves the effectiveness. Section 6 performs an experimental evaluation. Finally, Section 7 concludes the paper.

2. Related work

2.1. Data heterogeneity in FL

Data heterogeneity, characterized by non-IID data across clients, poses a significant challenge in FL [28,29]. Early mitigation strategies focused on regularizing local training processes. For example, FedProx [30] employs a proximal term to constrain local updates near the global model, while Scaffold [31] uses control variates to mitigate client drift. However, these methods are often inadequate for severe distribution shifts and fail to address system heterogeneity, such as varying client model architectures. Consequently, personalized FL (pFL) has emerged, prioritizing customized models for each client over a single global model [32]. For instance, FedBN [33] addresses feature shifts by maintaining client-specific Batch Normalization (BN) layers. Nevertheless, such approaches typically personalize only a limited portion of the model, rendering them less effective for significant domain gaps. While effective for statistical heterogeneity, they fall short in multi-domain scenarios where feature spaces diverge substantially.

2.2. Multi-domain and domain generalization in FL

In the realm of FL, the domain shift problem persists as a formidable challenge. Specifically, client data frequently inhabit distinct and non-overlapping domains, such as photographic images, hand-drawn sketches, and animated cartoons. This divergence starkly contrasts with the assumptions of standard FL protocols, which presuppose homogeneous data distributions across participants. Federated Domain Generalization (FedDG) [34] has emerged as a crucial research area to mitigate this issue. Current methodologies in FedDG predominantly encompass two categories: invariant representation learning, which endeavors to harmonize cross-domain feature distributions through statistical alignment techniques, and KD frameworks like MUFTI [35]. The latter approach orchestrates collaborative training of student models by leveraging domain-specific teacher models.

Although these established FedDG methods are theoretically promising, they are still hindered by significant limitations. Representation learning typically requires the exchange of sensitive domain-specific metadata, which directly violates the fundamental privacy protection principles of FL. Besides, methods based on a single KD typically rely on constructing generalized representations from sparse client datasets, making it difficult to effectively bridge substantial semantic and visual differences between different domains.

Concurrently, the integration of large-scale pre-trained models, such as Vision Transformers (ViT) [36] and CLIP [26], into FL architectures has instigated a wave of innovative techniques. Prompt-tuning methodologies like PromptFL [37] and adaptive prompt frameworks such as FedAPT [38] have been developed to optimize lightweight parameters, thereby reducing communication overhead in FL. However, these advancements are constrained by inherent bottlenecks: architectural homogeneity assumptions in existing frameworks fail to adequately address system heterogeneity, including variations in client-side computational architectures and hardware capabilities. For instance, FedCLIP [39] relies on manually crafted text descriptions for alignment, which underutilizes CLIP's intrinsic cross-modal generalization capabilities and restricts its adaptability to diverse domains.

2.3. KD

KD has gained traction for enabling model compression, cross-device collaboration, and personalized adaptation in FL. Approaches like FedKD [40] capitalizes on unlabeled and cross-domain public data to facilitate knowledge transfer, aiming to rectify data imbalance and safeguard privacy in FL. Nevertheless, its effectiveness hinges on the availability of ample public data, and in scenarios with limited resources, it often struggles to achieve optimal generalization. FedGKD [41] is designed to counteract the client drift issue prevalent in heterogeneous FL environments by leveraging historical global model knowledge to guide local training. Despite its focus on non-IID data, it falls short in fully accommodating the intricate data heterogeneity inherent in multi-domain settings, particularly in adapting to diverse feature and label distributions. FedGen [42] employs data-free KD via a server-side generator to synthesize features from client updates. However, as highlighted in recent studies, synthesizing high-fidelity features solely from model parameters is inherently unstable in high-dimensional, multi-domain spaces. The generator often fails to capture intricate inter-domain shifts, resulting in the synthesis of noisy or ambiguous features. When distilled back to clients, this noisy knowledge can mislead local training and degrade the discriminative capability of student models, particularly when domain discrepancies are severe.

To address these limitations, we propose FedCLIP-Distill, an advanced FL framework that leverages a pre-trained CLIP model as a universal semantic anchor. By distilling cross-modal representations from extensive image-text alignments into heterogeneous client models, this framework empowers federated participants to learn robust visual-semantic embeddings. This methodology effectively bridges domain gaps without necessitating the disclosure of explicit domain-specific details or the sharing of raw data. Leveraging CLIP's pre-trained cross-domain semantic knowledge as supervisory signals, FedCLIP-Distill grounds local training processes within a generalized knowledge space. Instead of relying on traditional parameter synchronization, it achieves alignment at the embedding level, ensuring compatibility with diverse client architectures. Besides, our model comprehensively mitigates domain transfer by leveraging the generalized visual semantic mapping of CLIP, addresses data heterogeneity through dual-domain KD, and adapted model heterogeneity through CRD, achieving privacy-protecting generalization without exchanging domain-specific metadata or raw data.

3. Preliminaries

3.1. CLIP

CLIP is a groundbreaking framework in the realm of multi-modal learning, specifically designed for image-text pre-training [26]. Given a collection of $(image, text)$ pairs denoted as $\mathcal{D} = \left\{\left(x_m, y_m\right)\right\}_{m=1}^{|\mathcal{D}|}$, CLIP performs a pivotal image-text alignment task. At its core, it involves optimizing the joint embedding space to maximize the cosine similarity between corresponding image-text pairs while simultaneously minimizing the similarity of non-matching pairs. The CLIP architecture consists of two fundamental components: a visual encoder f^i and a text encoder f^t. The visual encoder and text encoder respectively transform the input images and text into feature embeddings I_m and T_m. As shown in the following formula:

$$I_m = f^i(x_m), \quad T_m = f^t(y_m) \tag{1}$$

Next, all the embeddings are normalized using the l_2 norm. To achieve the goal of image-text alignment, CLIP leverages a symmetric cross-entropy loss [43]. The key idea behind this loss function is to maximize the similarity between the paired image and text embeddings. When the image embedding I_m is taken as the anchor, the image-to-text contrastive loss L_I is formulated as follows:

$$\mathcal{L}_I = -\log \frac{\exp(I_m \cdot T_m/\tau)}{\sum_{b=1}^{|\mathcal{B}|} \exp(I_m \cdot T_b/\tau)}, \tag{2}$$

where τ is a learnable temperature parameter that scales the distribution of the similarity scores, $\cdot$ denotes the dot product to measure the similarity and $\mathcal{B}$ is mini-batch.

In a symmetric manner, CLIP formulates the text-to-image contrastive loss. When the text embedding T_m serves as the anchor, the text-to-image contrastive loss L_T is given by:

$$\mathcal{L}_T = -\log \frac{\exp(T_m \cdot I_m/\tau)}{\sum_{b=1}^{|\mathcal{B}|} \exp(T_m \cdot I_b/\tau)}. \tag{3}$$

Finally, the total loss of the CLIP model, which balances the image-to-text and text-to -image alignment objectives, is defined as the average of these two contrastive losses:

$$\mathcal{L}_{CLIP} = \frac{1}{2}(\mathcal{L}_I + \mathcal{L}_T). \tag{4}$$

To adapt CLIP to KD tasks, a fundamental prerequisite involves establishing an effective mapping between CLIP-generated visual-text semantic similarity metrics and values that closely approximate classification logits. Within specific visual recognition tasks, these computed semantic similarity scores, which quantify the relationship between input images and class-relevant textual descriptions, can serve as viable substitutes for the logits produced by the final linear layer of standard deep neural network classifiers. This approach establishes a principled foundation for subsequent knowledge distillation probability computations.

3.2. KD

KD is a powerful model optimization technique, which transfers knowledge from a complex teacher model to a lightweight student model. Thereby it enhances the performance of the model while reducing computational costs. In KD, the logit output for an individual input is represented by $Z \in R^C$, where C denotes the number of distinct data categories, and the class probabilities are given by:

$$P_j = \frac{\exp(Z_j/T_{kd})}{\sum_{k=1}^{C} \exp(Z_k/T_{kd})}, \tag{5}$$

where Z_j does not represent the traditional logit, but the normalized semantic similarity between image embedding and class-specific text embedding for CLIP, and P_j represents the predicted probability of the j-th class. Unlike traditional softmax functions, KD achieves the transfer of soft targets by controlling the temperature T_{kd}, which helps mitigate the over-confidence problem in neural networks [44].

The distillation loss measures the discrepancy between the soft target distributions of the teacher and student models, commonly quantified using the Kullback-Leibler (KL) divergence:

$$\mathcal{L}_{KD} = KL(P_{\text{tea}} || P_{\text{stu}}) = \sum_{j=1}^{C} P_j^{\text{tea}} \log\left(\frac{P_j^{\text{tea}}}{P_j^{\text{stu}}}\right), \tag{6}$$

where P_j^{tea} and P_j^{tea} respectively denote the probability values for the j-th class of the teacher and student model outputs.

4. Methodology

4.1. Overview of FedCLIP-Distill

In the following section, we provide a detailed description of our proposed framework FedCLIP-Distill, as shown in Fig. 2 and Algorithm 1. In the Federated Multi-Domain Learning (FML) section, we introduce a collaborative training pipeline where a global server coordinates with

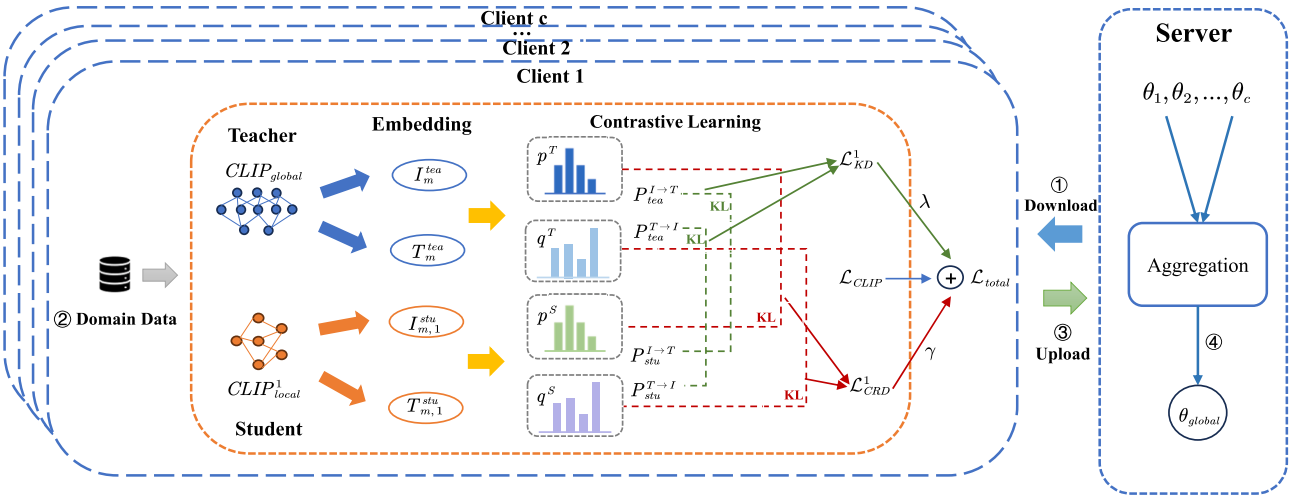

Fig. 2. Illustration of our FedCLIP-distill workflow. This workflow consists of four steps: 1. The server chooses a group of clients and sends the global model θ_{global} to them; 2. Each active client c initializes its local model θ_c based on the received global model, and then carries out local training using Dual-domain KD and Contrastive Relational Distillation; 3. The client sends the updated local model back to the server; 4. The server collects all the local models and combines them to form a new global model θ_{global}.

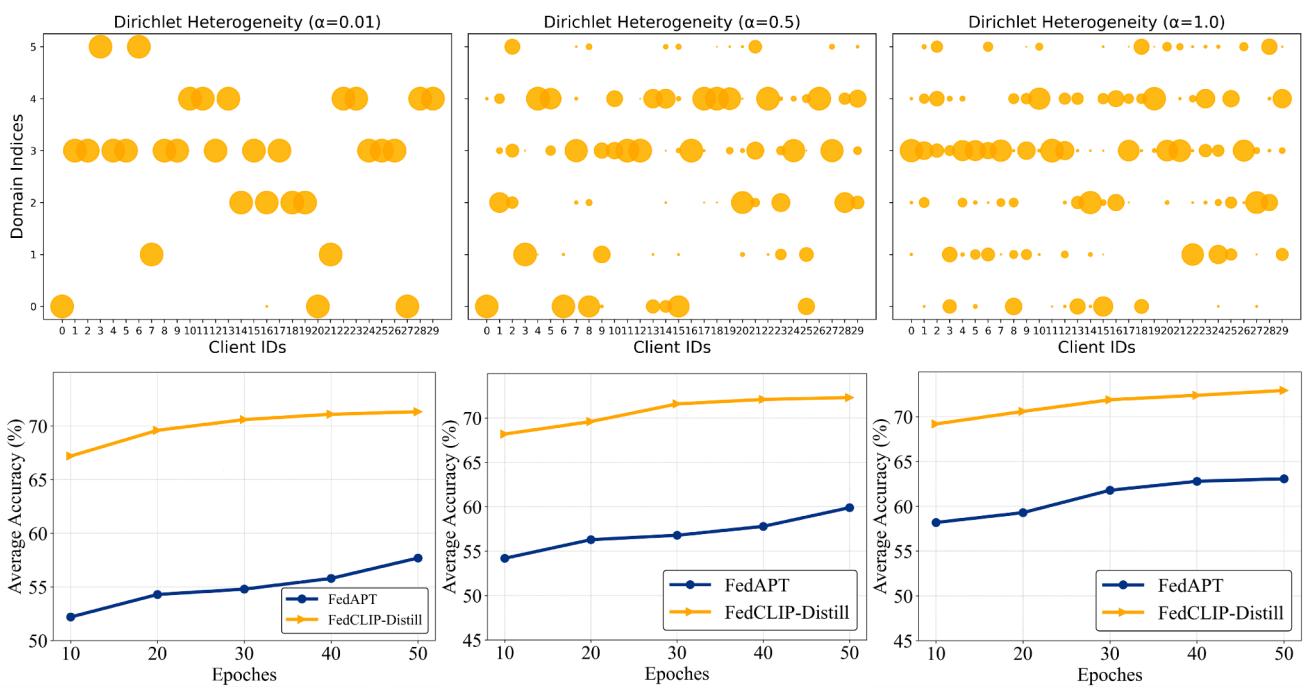

Fig. 3. Top: Visualization of statistical heterogeneity across clients in the DomainNet dataset with varying α values. Bottom: Performance of FedAPT and FedCLIP-Distill on DomainNet under different α setting.

local edge devices to train domain-specific student models while aggregating knowledge from a pre-trained global teacher model. Subsequently, the Dual-Domain KD section elaborates on a bidirectional knowledge transfer mechanism. This mechanism ensures that knowledge flows both from the global model to local models for generalization and from local models back to the global model for domain adaptation, effectively bridging the gaps between diverse data domains. To further enhance the knowledge transfer efficiency and model performance, we also incorporate Contrastive Relational Distillation(CRD), which focuses on refining the relational knowledge between data samples. Together, these components enable FedCLIP-Distill to achieve optimal cross-domain feature alignment, efficient knowledge transfer for enhanced generalization, and effective federated training under non-IID data conditions.

4.2. Federated multi-domain learning

In traditional FL frameworks, data heterogeneity and multi-domain scenarios pose significant challenges. Local models tend to overfit to domain-specific data distributions, resulting in poor generalization across diverse domains and subpar overall performance. To mitigate these issues, we introduce a method named FML.

Our framework uses a client-server architecture, with a global server and distributed edge clients. The global server holds a pre-trained global teacher model $CLIP_{global}$, trained on data from multiple domains. Each edge client c has its own domain-specific data D^c and runs a local student model $CLIP^c_{local}$. During training, the global server sends the $CLIP_{global}$ model to all clients. Then, each client c adjusts its local model using its own data.

After local training, clients send their updated model parameters to the server. The server combines these local models using weighted averaging to get the new global model:

$$\theta_{global} = \sum_{c=1}^{C} w_c \theta_c, \quad w_c = \frac{|D^c|}{\sum_{c=1}^{C} |D^c|}, \tag{7}$$

where C is the total number of clients, w_c is the weight of client c based on its data size, and θ_c is the parameters of the local model on client c. This process repeats until the model converges.

4.3. Dual-domain KD

Even with the collaborative training in FML, differences between domains still challenge model performance. We tackle this challenge through Dual-Domain KD, a bidirectional knowledge transfer framework designed to optimize information exchange.

For an image x_m^c and text y_m^c in domain c, the global teacher model $CLIP_{global}$ creates visual embedding I_m^{tea} and textual embedding T_m^{tea}, while the local student model $CLIP_{local}^c$ generates $I_{m,c}^{stu}$ and $T_{m,c}^{stu}$.

In the global-to-local distillation phase, we implement a visual-language alignment regularization to mitigate the semantic drift caused by local data heterogeneity. Since local clients typically hold non-IID data with distinct domain styles, updating the local visual encoder solely on private data risks decoupling the learned visual features from the generalized textual semantics. To counteract this, we utilize the pre-trained CLIP teacher as a semantic anchor and enforce the student model to replicate the teacher's cross-modal matching probability distributions. By minimizing the KL divergence between the student's and teacher's image-text similarity scores, we constrain the student's visual embedding space to remain strictly aligned with the teacher's stable and semantically rich textual embedding space. This regularization mechanism effectively prevents the student model from overfitting to biased local visual patterns, thereby ensuring robust semantic consistency across heterogeneous clients. The loss for aligning image to text is:

$$\mathcal{L}_{KD-IT} = KL(P_{tea}^{I\to T} || P_{stu}^{I\to T}), \tag{8}$$

where $P_{tea}^{I\to T}$ and $P_{stu}^{I\to T}$ are the soft probabilities from the global and local models respectively calculated by Eq. (5). A similar loss $\mathcal{L}_{KD-TI}$ defined for text to image alignment:

$$\mathcal{L}_{KD-TI} = KL(P_{tea}^{T\to I} || P_{stu}^{T\to I}). \tag{9}$$

In the local-to-global refinement, the local model sends its domain knowledge back to the global model. The refinement loss is:

$$\mathcal{L}_{KD-refine} = KL(P_{stu}^{I\to T} || P_{tea}^{I\to T}). \tag{10}$$

The dual-domain KD loss combines these losses:

$$\mathcal{L}_{KD}^c = \frac{1}{2}(\mathcal{L}_{KD-IT}^c + \mathcal{L}_{KD-TI}^c) + \mathcal{L}_{KD-refine}. \tag{11}$$

This bidirectional knowledge transfer mechanism empowers the model with enhanced cross-domain generalization capabilities.

4.4. CRD

To address the limitations of traditional KD in capturing relational structures among data samples, we introduce CRD [45]. While existing KD methods primarily focus on aligning individual sample representations, CRD explicitly models and transfers the relational knowledge embedded in the interactions between samples. This is particularly critical in FML scenarios, where preserving cross-sample relationships can significantly enhance generalization across diverse data distributions.

CRD constructs contrastive learning objectives that encourage the local model to preserve the same relational patterns as the global model. Given a mini-batch $\mathcal{B} = (x_m, y_m)_{m=1}^{|\mathcal{B}|}$ of image-text pairs from domain c, the global teacher model $CLIP_{global}$ and local student model $CLIP_{local}^c$ respectively generate visual-text embeddings $(I_m^{\mathrm{T}}, T_m^{\mathrm{T}})_{m=1}^{|\mathcal{B}|}$ and $(I_m^{\mathrm{S}}, T_m^{\mathrm{S}})_{m=1}^{|\mathcal{B}|}$. CRD constructs contrastive distributions to align relational patterns between the teacher and student models.

Algorithm 1 The framework of FedCLIP-Distill.

Input: Communication rounds T_{global}, local epochs T_{local}, batch size $\mathcal{B}$, learning rate η, Pre-trained global teacher model $\text{CLIP}_{\text{global}}$), Domain-specific client data D^c **Output**: global model θ_{global}

Server Execution::
Initialize global model parameters θ_{global}^0
for communication round $t = 1, ..., T_{global}$ **do**
  $M^t \leftarrow$ select a random set of $C \cdot M$ clients
  Send θ_{global}^t to all clients in M^t
  for each client $m \in M$, in parallel **do**
    $\theta_{t+1}^m \leftarrow$ ClientUpdate(m, θ_{global}^t)
    $\theta_{global}^{t+1} \leftarrow \sum_{m=1}^{M} w_c \theta_{\text{t+1}}^m$
  end
end

ClientUpdate(m, θ_{global}^t):
for $\theta_{\text{t}}^m \leftarrow \theta_{\text{global}}^t$
  for each local epoch $i = 1, ..., T_{local}$ **do**
    for each batch $b \in \mathcal{B}$ **do**
      $\theta_{t+1}^m \leftarrow \theta_t^m - \eta \cdot \nabla \mathcal{L}_{total}$ //Update the local model via minimizing $\mathcal{L}_{total}$.
    end
  end
end
Return θ_{t+1}^m back to server

For the mth image embedding v_k (from $\mathcal{B}$) as an anchor, we define image-to-text contrastive distributions for the teacher ($p_m^{\mathbf{T}}$) and student ($p_k^{\mathbf{S}}$) as:

$$p_m^{\mathbf{T}}[j] = \frac{\exp\left((I_m^{\mathbf{T}} \cdot T_j^{\mathbf{T}}) / \tau\right)}{\sum_{b=1}^{|\mathcal{B}|} \exp\left((I_m^{\mathbf{T}} \cdot T_b^{\mathbf{T}}) / \tau\right)}, \tag{12}$$

$$p_m^{\mathbf{S}}[j] = \frac{\exp\left((I_m^{\mathbf{S}} \cdot T_j^{\mathrm{S}}) / \tau\right)}{\sum_{b=1}^{|\mathcal{B}|} \exp\left((I_m^{\mathbf{S}} \cdot T_b^{\mathbf{S}}) / \tau\right)}, \tag{13}$$

where $j \in [1, 2, \ldots, |\mathcal{B}|]$ indexes text embeddings in $\mathcal{B}$, τ is temperature parameter for the teacher and student (the same temperature as Eq. (2), controlling distribution sharpness.

Symmetrically, for the mth text embedding s_k as an anchor, we define text-to-image contrastive distributions for the teacher ($q_m^{\mathbf{T}}$) and student ($q_m^{\mathbf{S}}$) as:

$$q_m^{\mathbf{T}}[j] = \frac{\exp\left((T_m^{\mathbf{T}} \cdot I_j^{\mathbf{T}}) / \tau\right)}{\sum_{b=1}^{|\mathcal{B}|} \exp\left((T_m^{\mathbf{T}} \cdot I_b^{\mathbf{T}}) / \tau\right)}, \tag{14}$$

$$q_m^{\mathbf{S}}[j] = \frac{\exp\left((T_m^{\mathbf{S}} \cdot I_j^{\mathrm{S}}) / \tau\right)}{\sum_{b=1}^{|\mathcal{B}|} \exp\left((T_m^{\mathbf{S}} \cdot I_b^{\mathbf{S}}) / \tau\right)}. \tag{15}$$

The CRD loss $\mathcal{L}_{\text{CRD}}^c$ aligns the teacher's and student's contrastive distributions via KL divergence. We aggregate losses for all anchors in $\mathcal{B}$:

$$\mathcal{L}_{\text{CRD}}^c = \frac{1}{|\mathcal{B}|} \sum_{m=1}^{|\mathcal{B}|} \left[\sum_{j=1}^{|\mathcal{B}|} \mathrm{KL}\big(p_m^{\mathbf{S}}[j] \parallel p_m^{\mathbf{T}}[j]\big) + \sum_{j=1}^{|\mathcal{B}|} \mathrm{KL}\big(q_m^{\mathbf{S}}[j] \parallel q_m^{\mathbf{T}}[j]\big) \right]. \tag{16}$$

Then we summarize the original FedCLIP task and distillation losses to jointly train a student model:

$$\mathcal{L}_{total} = \mathcal{L}_{CLIP} + \lambda \mathcal{L}^c_{KD} + \gamma \mathcal{L}^c_{CRD}, \tag{17}$$

where λ and γ are distillation loss weights to scale the magnitude.

5. Convergence analysis

Considering the heterogeneous case of FedCLIP-Distill, we prove the convergence of our algorithm under well-defined mathematical assumptions.

5.1. Assumptions

Assumption 1. (Lipschitz Continuity): The loss function $\mathcal{L}^c_{total}$ is L-smooth, meaning that its gradient is Lipschitz continuous. For any θ_1, θ_2, we have the following expressions:

$$\begin{aligned}\mathcal{L}^c_{total}(\theta_2) \le \mathcal{L}^c_{total}(\theta_1) \\ + \nabla \mathcal{L}^c_{total}(\theta_1)^T(\theta_2 - \theta_1) + \frac{L}{2}\|\theta_2 - \theta_1\|_2^2.\end{aligned} \tag{18}$$

Assumption 2. (Bounded Gradient Varience): The variance of local gradients across clients is bounded by σ^2. Formally, let $\mathcal{L}_{\text{total}}(\theta) = \sum_{c=1}^{C} w_c \mathcal{L}^c_{\text{total}}(\theta)$, then:

$$\mathbb{E}_c\left[\left\|\nabla \mathcal{L}^c_{\text{total}}(\theta) - \nabla \mathcal{L}_{\text{total}}(\theta)\right\|^2\right] \le \sigma^2. \tag{19}$$

Assumption 3. (Smoothness of KD Losses): The dual-domain KD loss $\mathcal{L}^c_{KD}$ and contrastive relational distillation loss $\mathcal{L}^c_{CRD}$ are β-smooth. For $\mathcal{L}^c_{KD}$, for any θ_1, θ_2:

$$\begin{aligned}\mathcal{L}^c_{KD}(\theta_2) \le \mathcal{L}^c_{KD}(\theta_1) \\ + \nabla \mathcal{L}^c_{KD}(\theta_1)^T(\theta_2 - \theta_1) + \frac{\beta}{2}\|\theta_2 - \theta_1\|_2^2.\end{aligned} \tag{20}$$

Similarly, for $\mathcal{L}^c_{CRD}$:

$$\begin{aligned}\mathcal{L}^c_{CRD}(\theta_2) \le \mathcal{L}^c_{CRD}(\theta_1) \\ + \nabla \mathcal{L}^c_{CRD}(\theta_1)^T(\theta_2 - \theta_1) + \frac{\beta}{2}\|\theta_2 - \theta_1\|_2^2.\end{aligned} \tag{21}$$

5.2. Lemma

Lemma 1. *(Federated Gradient Error Bound): The error between the global gradient and the aggregated local gradients is bounded as:*

$$\left\|\nabla \mathcal{L}_{total}(\theta) - \sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{total}(\theta)\right\| \le \frac{\sigma}{\sqrt{C}}. \tag{22}$$

Proof. Proof of Lemma 1: By the definition of variance and the bounded gradient variance assumption (Assumption 2), we can use the Cauchy-Schwarz inequality and the properties of expectation to derive this bound. For the detailed proof process, please refer to Appendix A. □

Lemma 2. *(Contraction of KD): The dual-domain and contrastive distillation losses enforce parameter similarity between global and local models. For $\lambda, \gamma > 0$, there exists $P \in (0, 1)$ such that:*

$$\mathbb{E}\left[\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2\right] \le P\mathbb{E}\left[\left\|\theta^c_t - \theta_t\right\|^2\right]. \tag{23}$$

Proof. Proof of Lemma 2: Using the smoothness of the KD losses (Assumption 3), we can expand the expected value of $\|\theta^c_{t+1} - \theta_{t+1}\|^2$. By substituting the expressions for θ^c_{t+1} and θ_{t+1} in terms of θ^c_t and θ_t, and using the properties of expectation and the loss function smoothness, we can show that the sequence of the expected squared differences contracts with a factor P. For the detailed proof process, please refer to Appendix B. □

5.3. Theorem

Theorem 1. *(Convergence of FedCLIP-Distill): Under Assumptions 1 – 3, if the learning rate η_t satisfies $\sum_{t=0}^{\infty} \eta_t = \infty$ and $\sum_{t=0}^{\infty} \eta_t^2 < \infty$, then the sequence $\{\theta_t\}$ generated by the FedCLIP-Distill converges to a stationary point θ^*, as shown in the following formula:*

$$\lim_{t\to\infty} \mathbb{E}\left[\left\|\nabla \mathcal{L}_{total}(\theta_t)\right\|^2\right] = 0. \tag{24}$$

Proof. Proof of Theorem 1: In FL, we have

$$\theta_{t+1} - \theta_t = -\eta_t \sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{\text{total}}(\theta^c_t) \tag{25}$$

to update global parameter. Substitute it into the Assumption 1 inequality:

$$\begin{aligned}\mathcal{L}_{\text{total}}(\theta_{t+1}) \le \mathcal{L}_{\text{total}}(\theta_t) - \eta_t \nabla \mathcal{L}_{\text{total}}(\theta_t)^T \sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{\text{total}}(\theta^c_t) \\ + \frac{L\eta_t^2}{2}\left\|\sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{\text{total}}(\theta^c_t)\right\|^2,\end{aligned} \tag{26}$$

let $\epsilon_t = \sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{\text{total}}(\theta^c_t) - \nabla \mathcal{L}_{\text{total}}(\theta_t)$. By Lemma 1, $\|\epsilon_t\| \le \frac{\sigma}{\sqrt{C}}$. And we rewrite $\sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{\text{total}}(\theta^c_t)$ as $\nabla \mathcal{L}_{\text{total}}(\theta_t) + \left(\sum_{c=1}^{C} w_c \nabla \mathcal{L}^c_{\text{total}}(\theta^c_t) - \nabla \mathcal{L}_{\text{total}}(\theta_t)\right)$.

So we can rewrite inequality (26) as:

$$\begin{aligned}\mathcal{L}_{\text{total}}(\theta_{t+1}) \le \mathcal{L}_{\text{total}}(\theta_t) - \eta_t\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2 - \eta_t \nabla \mathcal{L}_{\text{total}}(\theta_t)^T \epsilon_t \\ + \frac{L\eta_t^2}{2}\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t) + \epsilon_t\right\|^2.\end{aligned} \tag{27}$$

Next, taking the expectation of both sides with respect to the randomness in data:

$$\begin{aligned}\mathbb{E}\left[\mathcal{L}_{\text{total}}(\theta_{t+1})\right] \le \mathbb{E}\left[\mathcal{L}_{\text{total}}(\theta_t)\right] - \eta_t \mathbb{E}\left[\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2\right] \\ - \eta_t \mathbb{E}\left[\nabla \mathcal{L}_{\text{total}}(\theta_t)^T \epsilon_t\right] + \frac{L\eta_t^2}{2}\mathbb{E}\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t) + \epsilon_t\right\|^2,\end{aligned} \tag{28}$$

In the above formula, $\mathbb{E}\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t) + \epsilon_t\right\|^2$ can be expanded as $\mathbb{E}\left[\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2 + 2\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|\left\|\epsilon_t\right\| + \left\|\epsilon_t\right\|^2\right]$. Since $\mathbb{E}[\epsilon_t] = 0$ (by the definition of ϵ_t and the property of expectation), we can obtain $\mathbb{E}\left[\nabla \mathcal{L}_{\text{total}}(\theta_t)^T \epsilon_t\right] = 0$. Also, we can get the following inequality through Lemma 1:

$$\begin{aligned}&\mathbb{E}\left[\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2 + 2\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|\left\|\epsilon_t\right\| + \left\|\epsilon_t\right\|^2\right] \\ &\le \mathbb{E}\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2 + 2\frac{\sigma}{\sqrt{c}}\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\| + \frac{\sigma^2}{\sqrt{c}}.\end{aligned} \tag{29}$$

Then substitute it into the inequality (28):

$$\begin{aligned}\mathbb{E}\left[\mathcal{L}_{\text{total}}(\theta_{t+1})\right] \le \mathbb{E}\left[\mathcal{L}_{\text{total}}(\theta_t)\right] \\ - \eta_t\left(1 - \frac{L\eta_t}{2}\right)\mathbb{E}\left[\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2\right] + \mathcal{O}\left(\eta_t^2 + \frac{\eta_t}{\sqrt{C}}\right).\end{aligned} \tag{30}$$

Since $\mathcal{L}_{\text{total}}(\theta)$ is bounded below, and we know that $\sum_{t=0}^{T} \mathcal{O}(\eta_t^2 + \frac{\eta_t}{\sqrt{C}})$ converges when $\sum_{t=0}^{\infty} \eta_t^2 < \infty$. Because $\sum_{t=0}^{\infty} \eta_t = \infty$ and $\sum_{t=0}^{\infty} \eta_t^2 < \infty$, we must have:

$$\sum_{t=0}^{\infty} \eta_t\left(1 - \frac{L\eta_t}{2}\right)\mathbb{E}\left[\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2\right] < \infty. \tag{31}$$

Since $\eta_t\left(1 - \frac{L\eta_t}{2}\right) > 0$ for small enough η_t (and we can always choose the learning rate sequence to satisfy this condition), and $\sum_{t=0}^{\infty} \eta_t = \infty$, we conclude that:

$$\lim_{t\to\infty} \mathbb{E}\left[\left\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\right\|^2\right] = 0. \tag{32}$$

□

Table 1
Comparison results on DomainNet and Office-Caltech10 datasets. The best results are shown in **bold**.

Methods	DomainNet (DN)							Office-Caltech10 (OC)				
	c	i	p	q	r	s	Avg	a	w	d	c	Avg
FedAvg[22]	87.73	70.38	83.24	41.06	93.15	83.97	76.59	96.35	96.35	93.55	94.67	95.23
FedProx[30]	87.80	70.34	83.15	40.91	93.09	84.00	76.55	96.55	96.55	96.77	94.89	96.19
FedProto[50]	87.27	70.24	82.33	37.34	92.60	83.11	75.48	97.35	97.83	98.54	94.67	97.10
FedBN[33]	88.23	69.97	83.70	41.26	93.32	82.78	76.54	97.40	96.55	93.55	95.11	95.65
FedCLIP[39]	81.29	58.36	77.44	28.88	89.49	76.29	68.63	96.75	96.74	93.55	94.67	95.43
PromptFL[37]	88.05	65.97	80.19	42.03	89.81	80.24	74.38	97.92	98.45	97.65	94.22	97.06
FedAPT[38]	87.53	66.06	81.29	57.51	91.68	82.50	77.76	96.88	96.55	**100.00**	**96.00**	97.36
FedPGP[51]	79.34	66.12	53.76	45.19	78.34	83.12	67.65	97.92	98.28	**100.00**	96.00	98.05
FedOTP[52]	78.12	65.73	50.16	43.16	63.14	80.94	63.54	96.88	**100.00**	**100.00**	95.11	98.00
FedCLIP-Distill	**90.15**	**71.50**	**84.42**	**58.34**	**93.45**	**85.16**	**80.50**	**98.44**	**100.00**	**100.00**	95.56	**98.50**

6. Experients and analysis

6.1. Experimental Setting

To validate the effectiveness of our approach, we conduct evaluations on domain adaptation tasks using the Office-Caltech10 [46] and DomainNet [24] benchmarks.

Datasets We utilize two well-known datasets tailored for domain adaptation research: 1) Office-Caltech10: this dataset comprises four domains, namely amazon(a), caltech(c), dslr(d), and webcam(w). It features 10 overlapping object categories between the Office-31 and Caltech-256 datasets. The number of images per domain ranges from at least 100 to at most 800. Given the relatively small-scale nature of the dataset, each domain is treated as an individual client in our experiments. 2) DomainNet: a large-scale multi-domain dataset, it includes six domains: clipart(c), infograph(i), painting(p), quickdraw(q), real(r), and sketch(s). Each domain contains approximately 33k-120k images, covering 345 object categories.

Setting To comprehensively validate our approach, we evaluate KD performance under two distinct architectural configurations: 1) Homogeneous architecture: the teacher and student models adopt the same architectural type(e.g., Vit-B/32 and Vit-B/16). 2) Heterogeneous architecture: in this setup, the teacher and student models have disparate architectures(e.g., Vit-B/32 and ResNet50).

Data Heterogeneity Based on prior research [47–49], we utilize the Dirichlet distribution to characterize heterogeneous data. Specifically, our approach leverages the Dirichlet distribution Dir(α), which incorporates a configurable concentration parameter α. As α diminishes, the level of data heterogeneity intensifies, effectively simulating more pronounced distribution disparities across datasets. As illustrated in Fig. 3, we simulate varying degrees of data heterogeneity on the DomainNet dataset by configuring the Dirichlet concentration parameter α to values of 0.01, 0.5, and 1.0. These specific values are chosen to evaluate the model across a comprehensive spectrum of distribution shifts: $\alpha = 0.01$ simulates an extreme non-IID scenario characterized by severe label skew, where clients hold samples from very few classes; $\alpha = 0.5$ represents a moderate level of heterogeneity commonly encountered in federated settings; and $\alpha = 1.0$ serves as a baseline for near-IID scenarios. In the $\alpha = 1.0$ setting, the distribution tends toward uniformity with clients possessing approximately similar class proportions, allowing us to verify that our method maintains robust performance even when data distributions are relatively balanced.

6.2. Compared methods

To comprehensively evaluate the efficacy of FedCLIP-Distill, we conduct comparisons against a diverse set of baselines ranging from classical heterogeneity-aware algorithms to state-of-the-art vision-language frameworks. We include traditional approaches such as FedAvg [22] as a standard baseline, FedBN [33] for addressing feature shift via local batch normalization, FedProx [30] for handling system heterogeneity through proximal regularization, and FedProto [50] for prototype-based semantic alignment. For CLIP-specific methods, we utilize CLIP-zeroshot as a non-learning baseline, alongside FedCLIP [39] and PromptFL [37], which represent standard prompt-based generalization strategies. Furthermore, we incorporate advanced heterogeneity-aware methods adapted for CLIP, including FedAPT [38] for covariate shift adaptation via dynamic parameter tuning, FedPGP [51] which employs Low-Rank Adaptation (LoRA) and prompt contrastive loss for balancing regularization and personalization, and FedOTP [52] which utilizes optimal transport for distribution alignment. Finally, these benchmarks are compared against our proposed FedCLIP-Distill, which synergistically integrates Dual-Domain KD and CRD to simultaneously mitigate data and model heterogeneity.

6.3. Implementation details

We implement all methods using PyTorch with the SGD optimizer for consistent training across datasets. For Office-Caltech10, we set a learning rate of 0.001 and batch size of 32, while DomainNet uses a learning rate of 0.01 and batch size of 256 to accommodate its larger scale and diversity. In federated training, we configure 50 global communication rounds (T_{global}) and 1 local training epoch (T_{local}) per round to balance convergence and communication efficiency. All experiments are completed with one GeForce RTX 4090 GPU.

6.4. Main results

Model Performance Evaluation. In the evaluation of domain adaptation performance on Office-Caltech10 and DomainNet datasets, we treated each domain as an independent client with the same class distribution but distinct domain-specific features. Our proposed FedCLIP-Distill, when compared with FedCLIP, PromptFL, and FedAPT, demonstrated significant advantages. The outcomes presented in Tables 1 and 2 substantiate our method's advantages in addressing generalization and semantic consistency:

Verification of Generalization Capability: As shown in Table 1, FedCLIP-Distill achieves the highest average accuracy on the DomainNet dataset (73.36%), outperforming FedAPT (73.34%) and PromptFL (72.66%). Notably, on domains with significant visual abstraction and distribution shifts, such as 'Quickdraw' (q) and 'Infograph' (i), our method demonstrates superior robustness. While FedAPT relies on adaptive parameter tuning which may overfit to local domain characteristics, FedCLIP-Distill utilizes the Dual-Domain KD mechanism. This effectively transfers the teacher's global perspective to local clients, preventing the model from becoming biased towards specific local visual styles and thereby enhancing its ability to generalize across heterogeneous domains.

Verification of Semantic Consistency: The effectiveness of our approach in maintaining semantic consistency is further evidenced by the results in highly heterogeneous settings (Table 2, $\alpha = 0.01$). In this extreme non-IID scenario, FedCLIP-Distill maintains an average accuracy of 71.33%,

Table 2
Accuracy comparison on DomainNet (DN) and Office-Caltech 10(OC) datasets under different heterogeneity levels (β). The best results are shown in **bold**.

β	Methods	DomainNet (DN)							Office-Caltech 10(OC)				
		c	i	p	q	r	s	Avg	a	w	d	c	Avg
0.01	FedAvg[22]$_{[AISTATS'17]}$	81.37	59.28	77.32	26.69	89.29	75.53	68.25	93.75	89.66	93.55	93.78	92.69
	FedProx[30]$_{[MLSys'20]}$	81.47	59.56	77.78	26.94	89.63	76.08	68.58	95.31	94.83	96.77	95.56	95.62
	FedProto[50]$_{[AAAI'22]}$	83.68	65.54	76.79	26.86	88.38	76.50	69.63	94.79	93.10	93.55	95.33	94.19
	FedBN[33]$_{[ICLR'21]}$	82.53	62.05	74.11	26.43	89.46	75.22	68.30	94.79	94.83	96.77	95.11	95.38
	FedCLIP[39]$_{[ICLR'23]}$	82.99	58.60	76.53	25.83	89.26	75.16	68.06	95.83	93.10	93.55	93.33	93.95
	PromptFL[37]$_{[TMC'24]}$	85.94	38.36	62.79	21.71	79.96	67.39	59.36	96.35	97.56	98.98	94.22	96.78
	FedAPT[38]$_{[AAAI'24]}$	82.69	39.69	59.31	21.60	77.34	65.59	57.70	97.91	98.28	**100.00**	**96.44**	98.16
	FedPGP[51]$_{[ICML'24]}$	78.13	64.13	50.12	28.65	66.54	76.23	60.63	97.92	98.28	**100.00**	**96.44**	98.16
	FedOTP[52]$_{[CVPR'24]}$	77.09	63.17	48.60	28.11	61.80	74.63	58.90	95.31	98.28	**100.00**	94.67	97.07
	FedCLIP-Distil **[Ours]**	**87.07**	**65.15**	**78.85**	**46.31**	**90.50**	**80.18**	**74.68**	**98.44**	**100.00**	**100.00**	94.67	**98.28**
0.5	FedAvg[22]$_{[AISTATS'17]}$	82.06	59.23	77.27	28.20	89.50	76.39	68.78	94.79	92.83	95.77	94.67	94.52
	FedProx[30]$_{[MLSys'20]}$	83.49	62.73	79.46	28.94	90.96	78.75	70.72	95.83	94.83	96.77	95.11	95.64
	FedProto[50]$_{[AAAI'22]}$	84.25	67.24	80.16	28.63	90.57	78.08	71.49	95.31	93.41	93.67	96.10	94.82
	FedBN[33]$_{[ICLR'21]}$	84.68	65.65	79.59	28.17	90.74	78.61	71.24	95.83	94.83	96.77	95.56	95.75
	FedCLIP[39]$_{[ICLR'23]}$	82.66	58.60	76.68	25.43	89.38	75.37	68.02	95.92	94.23	94.12	95.33	94.90
	PromptFL[37]$_{[TMC'24]}$	87.86	53.00	64.90	22.54	83.04	69.78	63.52	97.39	98.27	**100.00**	95.11	97.69
	FedAPT[38]$_{[AAAI'24]}$	83.85	49.01	61.72	18.20	80.25	66.41	59.91	97.32	97.85	**100.00**	95.33	97.63
	FedPGP[51]$_{[ICML'24]}$	78.73	65.16	51.45	28.93	66.84	76.98	61.35	97.92	98.28	**100.00**	**97.33**	98.38
	FedOTP[52]$_{[CVPR'24]}$	77.59	63.87	49.34	28.71	62.10	75.33	59.49	95.31	**100.00**	**100.00**	92.89	97.05
	FedCLIP-Distill **[Ours]**	**89.19**	**70.34**	**83.82**	**53.37**	**92.31**	**84.89**	**78.99**	**98.44**	**100.00**	**100.00**	95.11	**98.39**
1	FedAvg[22]$_{[AISTATS'17]}$	83.96	59.76	77.93	28.74	89.86	76.78	69.51	95.83	94.83	96.77	95.56	95.75
	FedProx[30]$_{[MLSys'20]}$	84.18	67.24	80.16	29.40	90.61	79.05	71.77	96.23	94.96	97.01	95.66	95.97
	FedProto[50]$_{[AAAI'22]}$	84.12	67.13	80.16	29.57	90.56	79.01	71.76	95.72	93.67	93.98	96.33	94.93
	FedBN[33]$_{[ICLR'21]}$	85.56	67.65	80.35	28.54	91.42	78.30	71.97	97.40	96.55	96.77	96.89	96.90
	FedCLIP[39]$_{[ICLR'23]}$	83.26	58.98	77.05	25.86	89.61	75.65	68.40	96.34	95.72	95.67	94.91	95.66
	PromptFL[37]$_{[TMC'24]}$	87.83	55.97	64.84	23.09	82.87	70.54	64.19	95.83	98.28	**100.00**	94.67	97.20
	FedAPT[38]$_{[AAAI'24]}$	84.58	55.09	65.05	20.54	83.13	70.11	63.08	97.92	98.28	**100.00**	96.44	98.16
	FedPGP[51]$_{[ICML'24]}$	78.87	65.53	51.65	28.99	67.02	77.04	61.52	97.92	98.28	**100.00**	**97.33**	98.38
	FedOTP[52]$_{[CVPR'24]}$	77.82	64.07	49.76	28.92	62.53	75.93	59.84	95.31	**100.00**	**100.00**	93.78	97.27
	FedCLIP-Distill **[Ours]**	**89.56**	**70.96**	**83.70**	**57.20**	**92.97**	**84.83**	**79.87**	**99.48**	**100.00**	**100.00**	96.89	**99.09**

Table 3
Model performance across heterogeneous architectures on DomainNet dataset.

$\mathcal{D}$	T:Vit-B/32 S:ResNet50	T:Vit-B/32 S:ResNet101	T:Vit-L/14 S:Vit-B/32	T:Vit-L/14 S:Vit-B/16
c	70.19	76.75	80.77	84.18
i	56.28	60.31	57.88	66.55
p	68.74	74.76	76.34	79.33
q	11.80	18.31	22.94	28.54
r	87.16	89.03	88.86	90.73
s	65.98	73.45	74.46	78.28
avg	60.03	65.44	66.88	71.27

Table 4
Model performance across heterogeneous architectures on Office-Caltech10 dataset.

$\mathcal{D}$	T:Vit-B/32 S:ResNet50	T:Vit-B/32 S:ResNet101	T:Vit-L/14 S:Vit-B/32	T:Vit-L/14 S:Vit-B/16
a	96.88	97.92	97.40	97.40
w	96.55	98.28	100.00	100.00
d	96.77	100.00	100.00	96.77
c	95.11	96.00	95.56	96.44
avg	96.33	98.05	98.24	97.65

significantly surpassing FedAPT (61.80%) and PromptFL (59.36%). Existing methods like PromptFL optimize prompts based on local data, which can lead to semantic drift when local data is class-imbalanced or noisy. In contrast, our CRD module enforces the alignment of feature relationships between the student and the frozen CLIP teacher. This ensures that the student's feature embedding space remains semantically consistent with the robust, pre-trained text-image alignment of CLIP, even when trained on fragmented local data.

Table 5
Ablation study. The experiment is conducted on DomainNet and Office-Caltech10 with $\alpha = 0.5$, using avg accuracy as the evaluation metric.

FedCLIP	Dual-domain KD	CRD	DomainNet (Acc)	Office-Caltech10 (Acc)
✔	✗	✗	68.02	94.90
✔	✔	✗	70.69	97.32
✔	✗	✔	69.49	95.84
✔	✔	✔	74.68	98.39

Data Heterogeneity Analysis. To introduce more complex heterogeneity, we further split each domain in DomainNet into multiple sub-clients. Specifically, we sample $\boldsymbol{\pi}_c \sim \mathrm{Dir}(\alpha \mathbf{1}_K)$ (where K is the number of sub-clients per domain, set to 5 in our experiments) and allocate a proportion $\pi_{c,i}$ of instances with label c to the ith sub-client within each domain [45]. A smaller α value leads to greater discrepancies in data distribution among sub-clients and more severe class imbalance. Through this process, we obtain a total of 30 sub-clients (6 domains × 5 sub-clients per domain) for more comprehensive evaluation. The results are shown in Tables 2 and 4. It is evident that our model demonstrates superior performance. Compared to FedCLIP, which is more sensitive to category distribution differences, our method's KD mechanism helps in transferring and aligning category-level knowledge across sub-clients. Although PromptFL offers flexibility in prompt-based adaptation, it falls short of FedCLIP-Distill's performance in scenarios characterized by simultaneous domain and category heterogeneity. FedAPT, though efficient, faces difficulties in handling the combined heterogeneities as effectively as our proposed method, which is designed to mitigate such complex divergence through cross-domain and cross-category KD.

Model Heterogeneity Analysis. When it comes to the situation where the teacher and student models are architecturally heterogeneous, the results in Table 3 indicate that our method shows

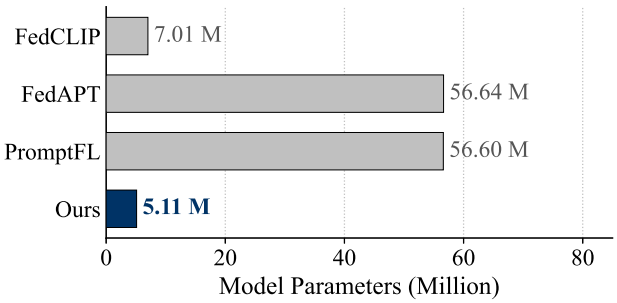

Fig. 4. Visualization of different model parameter.

significant advantages over previous methods and effectively enhances the lightweight student model.

Model Parameter Analysis. As illustrated in Fig. 4, the FedClip-Distill model demonstrates a notable advantage in parameter efficiency. This lightweight architecture is attributed to the KD framework, which compresses the pre-trained CLIP model by transferring essential image-text knowledge from the teacher model to a compact student model. By prioritizing the extraction of task-relevant representations rather than preserving all pre-trained parameters, we eliminate the redundancy associated with full-parameter fine-tuning and the complexity involved in extensive prompt engineering or adaptive parameter configurations. The reduced parameter count not only minimizes storage and computational demands during federated training but also enhances communication efficiency in distributed environments. This makes the model particularly well-suited for deployment on resource-constrained edge devices while maintaining competitive performance, thereby demonstrating its strengths in balancing model capacity, efficiency, and practical applicability.

Furthermore, we address the practical challenge of unknown and uncontrolled heterogeneity often encountered in real-world deployments. Unlike controlled experimental settings where heterogeneity is parameterized, real-world data distributions are typically opaque and volatile. FedCLIP-Distill informs mitigation strategies for such scenarios through its anchor-based design: the pre-trained CLIP teacher acts as a fixed semantic reference point, decoupling the model's robustness from the need to estimate local data distributions. The Dual-Domain KD and CRD mechanisms function as adaptive regularizers, where the corrective gradients naturally intensify in response to larger deviations caused by severe heterogeneity. This intrinsic adaptivity allows the framework to maintain semantic consistency and generalization capability across varying and unpredictable levels of client heterogeneity without requiring explicit prior knowledge or hyperparameter reconfiguration.

6.5. Parameter sensitivity analysis

In the FedCLIP-Distill framework, λ balances the contributions of the CLIP alignment loss (L_{CLIP}) and the KD loss (L^{c}_{KD}). A higher λ prioritizes transferring task-specific knowledge from the teacher model, while a lower λ emphasizes preserving CLIP's general text-image alignment. To investigate the joint impact of the KD weight (λ) and the contrastive regularization weight (γ) on model performance, we conduct a grid search analysis visualized in Fig. 5. We observe a distinct region of optimality centered at $\lambda = 0.05$ and $\gamma = 0.25$, where the model achieves a peak accuracy of 74.68%. It shows $\lambda = 0.05$ provides sufficient teacher guidance to transfer task-specific knowledge without compromising the student's generalizability, while $\gamma = 0.25$ effectively enhances intra-class compactness and inter-class separability via contrastive learning. Deviating from this region leads to performance degradation; specifically, lower values of λ or γ result in underfitting due to inadequate knowledge transfer or weak feature discrimination, whereas excessively high values induce over-regularization, suppressing the diversity of learned representations. Crucially, the broad high-accuracy area surrounding the peak demonstrates that our framework is robust to moderate hyperparameter perturbations, ensuring stable convergence in practical deployment.

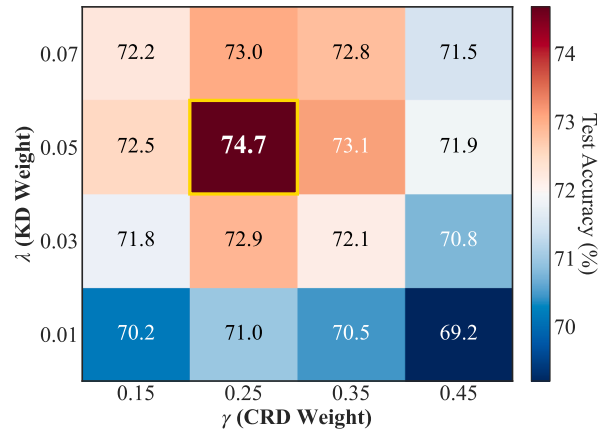

Fig. 5. Parameter sensitivity analysis. This is a combined parameter heatmap that can simultaneously visualize the interaction between the KD weight (λ) and the CRD weight (γ).

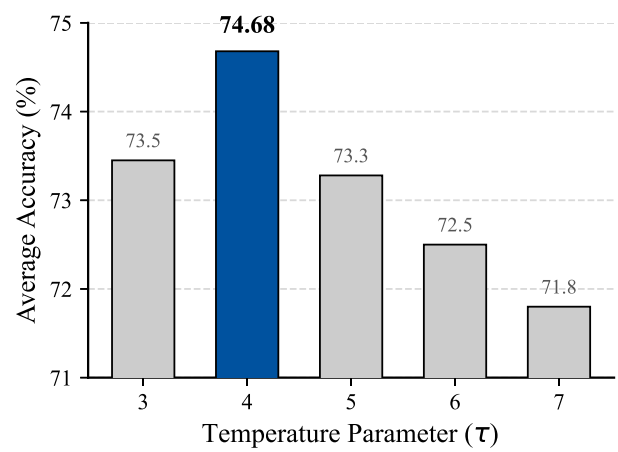

Fig. 6. Parameter sensitivity analysis. Experiments are conducted on DomainNet, with the teacher and student models in homogenous architecture.

Here, we conduct hyperparameter sensitivity on τ. According to our experimental setup, we took K = 5 temperatures and $\tau =$ [3.0, 4.0, 5.0, 6.0, 7.0], and evaluated the model on DomainNet using the teacher model and student model under the homogeneous architecture respectively. Finally, we select temperature $\tau = 4.0$. The results are shown in the Fig. 6.

6.6. Ablation studies

We conduct an ablation study to assess the effectiveness of different components in our proposed framework, comparing FedCLIP, FedCLIP with Dual-domain KD, and FedCLIP with Dual-domain KD plus CRD. As shown in Table 5, FedCLIP, which solely utilizes the CLIP model in a FL context, attains an average accuracy of 68.02% across multiple domains. By integrating Dual-domain KD into FedCLIP, the performance surges significantly to 70.69%, validating that distilling knowledge from diverse domains enables the student model to capture varied information and enhances its generalization ability. CRD enforces contrastive alignment of features across different domains, facilitating mutual learning and discrimination. This process prevents feature collapse and equips the model with stronger capabilities to distinguish between classes and domain-specific characteristics. These results validate that each of our

components is necessary and can significantly improve the performance of the model.

7. Conclusion and future work

In this article, we introduce FedCLIP-Distill, which is an innovative framework that integrates the KD mechanism into FL and utilizes pre-trained CLIP models for multi-domain scenarios. We designed a Dual-domain KD that combines multi-domain distillation to bridge the semantic gap across domains and combines CRD to enhance the discriminability of features. We not only proved the convergence of our model theoretically, but also verified the validity of the model through a large number of experiments. By optimally tuning hyperparameters to achieve the best balance, our framework has achieved outstanding performance, even when dealing with heterogeneous client data with different feature distributions and category differences. Compared with the baseline method, FedCLIP-Distill has significant improvements in both accuracy and domain generalization. It effectively extracts and aggregates cross-domain knowledge, enabling the student model to learn more robust and discriminative representations with fewer learnable parameters, thereby providing a promising solution for multi-domain FL tasks.

Limitations and Future Work: A limitation of our current approach is the requirement for manual hyperparameter tuning based on the dataset, making it difficult to adapt to scenarios where the data distribution on the client side changes dynamically. In large-scale client scenarios, embedded switching increases communication overhead, and weighted aggregation of data volume is prone to bias towards clients with large datasets. Future work could explore meta-learning based adaptive weighting schemes to dynamically adjust these hyperparameters during training.

CRediT authorship contribution statement

Yuankun Xia: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft; **Hui Wang:** Conceptualization, Resources, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing; **Yufeng Zhou:** Validation, Visualization, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A.

Proof. Proof of Lemma 1:

Let $\mathcal{L}_{\text{total}}(\theta) = \sum_{c=1}^{C} w_c \mathcal{L}_{\text{total}}^c(\theta)$ denote the global loss. For each client c, define the local-to-global gradient deviation:

$$\Delta_c(\theta) = \nabla \mathcal{L}_{\text{total}}^c(\theta) - \nabla \mathcal{L}_{\text{total}}(\theta). \tag{A.1}$$

So the aggregated local gradients are:

$$\begin{aligned}\sum_{c=1}^{C} w_c \nabla \mathcal{L}_{\text{total}}^c(\theta) &= \sum_{c=1}^{C} w_c \left[\nabla \mathcal{L}_{\text{total}}(\theta) + \Delta_c(\theta)\right] \\ &= \nabla \mathcal{L}_{\text{total}}(\theta) + \sum_{c=1}^{C} w_c \Delta_c(\theta).\end{aligned} \tag{A.2}$$

Next, we analyze the norm of the residual term $\sum_{c=1}^{C} w_c \Delta_c(\theta)$. By the Cauchy-Schwarz inequality, we have:

$$\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2 \le \left(\sum_{c=1}^{C} w_c\right) \cdot \left(\sum_{c=1}^{C} w_c \|\Delta_c(\theta)\|^2\right). \tag{A.3}$$

Since $\sum_{c=1}^{C} w_c = 1$, this reduces to:

$$\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2 \le \sum_{c=1}^{C} w_c \|\Delta_c(\theta)\|^2. \tag{33}$$

Then taking the expectation of both sides and using Assumption 2:

$$\mathbb{E}\left[\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2\right] \le \mathbb{E}\left[\sum_{c=1}^{C} w_c \|\Delta_c(\theta)\|^2\right]. \tag{A.4}$$

And by linearity of expectation:

$$\begin{aligned}\mathbb{E}\left[\sum_{c=1}^{C} w_c \|\Delta_c(\theta)\|^2\right] &= \sum_{c=1}^{C} w_c \mathbb{E}\left[\|\Delta_c(\theta)\|^2\right] \\ &\le \sum_{c=1}^{C} w_c \cdot \sigma^2 = \sigma^2.\end{aligned} \tag{A.5}$$

From Jensen's inequality, we can learn:

$$\mathbb{E}\left[\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2\right] \ge \left\|\mathbb{E}\left[\sum_{c=1}^{C} w_c \Delta_c(\theta)\right]\right\|^2, \tag{A.6}$$

and we have $\mathbb{E}\left[\sum_{c=1}^{C} w_c \Delta_c(\theta)\right] = 0$. So we have:

$$\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2 \le \sigma^2. \tag{A.7}$$

We rearrange inequality (A.2):

$$\nabla \mathcal{L}_{\text{total}}(\theta) - \sum_{c=1}^{C} w_c \nabla \mathcal{L}_{\text{total}}^c(\theta) = -\sum_{c=1}^{C} w_c \Delta_c(\theta). \tag{A.8}$$

Next taking norms and using the bound from inequality (A.8):

$$\begin{aligned}\left\|\nabla \mathcal{L}_{\text{total}}(\theta) - \sum_{c=1}^{C} w_c \nabla \mathcal{L}_{\text{total}}^c(\theta)\right\| &\le \sqrt{\mathbb{E}\left[\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2\right]} \\ &\le \sigma.\end{aligned} \tag{A.9}$$

In practice, the weights w_c are proportional to data sizes $|D^c|$. For simplicity, assume $w_c = \frac{1}{C}$. Then:

$$\left\|\sum_{c=1}^{C} w_c \Delta_c(\theta)\right\|^2 = \frac{1}{C^2}\left\|\sum_{c=1}^{C} \Delta_c(\theta)\right\|^2. \tag{A.10}$$

By the variance of sums:

$$\begin{aligned}\mathbb{E}\left[\left\|\sum_{c=1}^{C} \Delta_c(\theta)\right\|^2\right] &= \sum_{c=1}^{C} \mathbb{E}\left[\|\Delta_c(\theta)\|^2\right] \\ &\quad + 2\sum_{1\le i<j\le C} \mathbb{E}\left[\Delta_i(\theta)^T \Delta_j(\theta)\right].\end{aligned} \tag{A.11}$$

Since $\mathbb{E}[\Delta_c(\theta)] = 0$, the cross-terms vanish under independence, giving:

$$\mathbb{E}\left[\left\|\sum_{c=1}^{C}\Delta_c(\theta)\right\|^2\right] \le C\sigma^2. \tag{A.12}$$

Thus:

$$\mathbb{E}\left[\left\|\sum_{c=1}^{C}w_c\Delta_c(\theta)\right\|^2\right] \le \frac{C\sigma^2}{C^2} = \frac{\sigma^2}{C}. \tag{A.13}$$

Last, taking square roots, we can get:

$$\left\|\nabla\mathcal{L}_{\text{total}}(\theta) - \sum_{c=1}^{C}w_c\nabla\mathcal{L}^c_{\text{total}}(\theta)\right\| \le \frac{\sigma}{\sqrt{C}}. \tag{A.14}$$

□

Appendix B.

Proof. Proof of Lemma 2:

First, we start by expanding $\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2$:

$$\begin{aligned}\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2 &= \left\|\theta^c_{t+1} - \sum_{k=1}^{C}w_k\theta^k_t\right\|^2 \\ &= \left\|\left(\theta^c_{t+1} - \theta^c_t\right) + \left(\theta^c_t - \theta_t\right) + \left(\theta_t - \sum_{k=1}^{C}w_k\theta^k_t\right)\right\|^2.\end{aligned} \tag{B.1}$$

By the property of the squared norm $\|a+b+c\|^2 \le 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$ and $2|a^Tb| \le a^Ta + b^Tb$, we have:

$$\begin{aligned}&\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2 \\ &\le 3\left(\left\|\theta^c_{t+1} - \theta^c_t\right\|^2 + \left\|\theta^c_t - \theta_t\right\|^2 + \left\|\theta_t - \sum_{k=1}^{C}w_k\theta^k_t\right\|^2\right).\end{aligned} \tag{B.2}$$

Next we analysis the update of the local model, the local model θ^c_{t+1} is updated using gradient descent: $\theta^c_{t+1} = \theta^c_t - \eta\nabla\mathcal{L}^c_{\text{total}}(\theta^c_t)$. The squared norm of this update is:

$$\left\|\theta^c_{t+1} - \theta^c_t\right\|^2 = \eta^2\left\|\nabla\mathcal{L}^c_{\text{total}}(\theta^c_t)\right\|^2. \tag{B.3}$$

By Assumptions 1 and 3, the total loss $\mathcal{L}^c_{total}$ is L-smooth. For a minimizer θ^c_{t+1} of $\mathcal{L}^c_{total}$, the update satisfies:

$$\left\|\theta^c_{t+1} - \theta^c_t\right\|^2 \le \mu_1\left\|\theta^c_t - \theta_t\right\|^2, \tag{B.4}$$

where μ_1 is a constant related to η and L. Then analyze global aggregation term $\theta_t - \sum_{k=1}^{C}w_k\theta^k_t$. The global aggregation term measures deviation from federated averaging. By convexity of the squared norm and $\sum_{k=1}^{C}w_k = 1$:

$$\begin{aligned}\left\|\theta_t - \sum_{k=1}^{C}w_k\theta^k_t\right\|^2 &= \left\|\sum_{k=1}^{C}w_k(\theta_t - \theta^k_t)\right\|^2 \\ &\le \sum_{k=1}^{C}w_k\left\|\theta_t - \theta^k_t\right\|^2.\end{aligned} \tag{B.5}$$

Next, take the expectation of inequality (B.2):

$$\begin{aligned}&\mathbb{E}\left[\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2\right] \\ &\le 3\mathbb{E}\left[\left\|\theta^c_{t+1} - \theta^c_t\right\|^2 + \left\|\theta^c_t - \theta_t\right\|^2 + \left\|\theta_t - \sum_{k=1}^{C}w_k\theta^k_t\right\|^2\right],\end{aligned} \tag{B.6}$$

and substitute bounds from inequality (B.5) and (B.6):

$$\begin{aligned}&\mathbb{E}\left[\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2\right] \le 3(\mu_1\mathbb{E}\left[\left\|\theta^c_t - \theta_t\right\|^2\right] \\ &+ \mathbb{E}\left[\left\|\theta^c_t - \theta_t\right\|^2\right] + \sum_{k=1}^{C}w_k\mathbb{E}\left[\left\|\theta_t - \theta^k_t\right\|^2\right]).\end{aligned} \tag{B.7}$$

For non-IID data, assume $\mu_1 < \frac{1}{3}$. Define $\alpha = 3(\mu_1 + 1)$, and note $\sum_{k=1}^{C}w_k\mathbb{E}\left[\left\|\theta_t - \theta^k_t\right\|^2\right]$ is absorbed into the bound. Thus:$\mathbb{E}\left[\left\|\theta^c_{t+1} - \theta_{t+1}\right\|^2\right] \le P\mathbb{E}\left[\left\|\theta^c_t - \theta_t\right\|^2\right]$, where $P \in (0, 1)$. □

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